In 1850 Franz Nauck asked how many ways N queens can be placed on an N by N board so that no two attack, generalizing the eight queens question that Gauss had taken up. The puzzle is a window on the central problem of computer science. Completing a partial board is NP complete, counting its completions is #P complete, and every Monte Carlo method proposed for such counts comes down to a walk along a single staircase on the unit interval: the quantile function of the likelihood, whose integral is the Lorenz curve identity behind nested sampling and vertical likelihood.

This book sets out that observation as a program and marks its gaps. It proves that NP = RP is equivalent to polynomial time sampling and approximate counting of queens completions, and states as a conjecture when a swap chain on the energy levels mixes fast enough to deliver them; it restates promise RP = promise P, which implies P = RP, as the known hitting set problem on the quantile axis, which it does not claim to solve; it gives Yakowitz type bounds, higher order and stratified rules, and generative Bayesian computation as the machinery; and it reports exact computations on the queens: the density of states, the giant component of solutions, its persistence on conditioned boards, and the shattering depth of the bottom level sets.

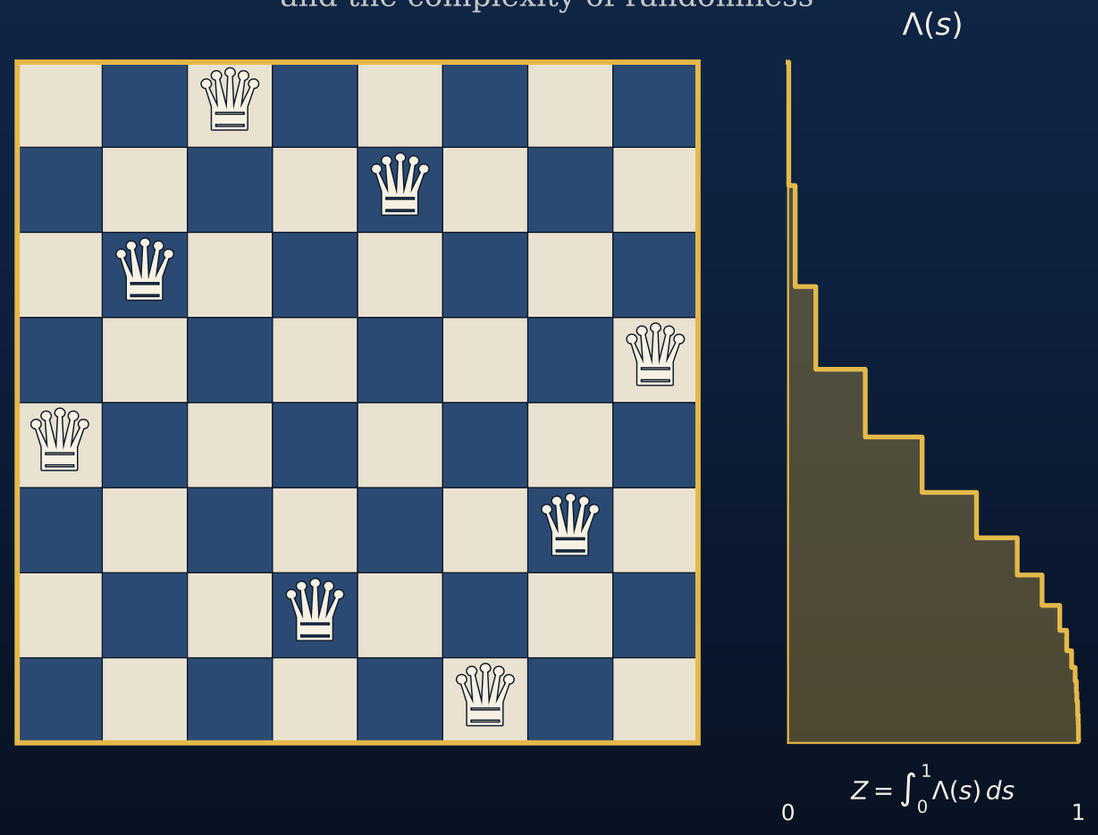
From Hilbert and Turing through Shannon and von Neumann to Jerrum, Sinclair and Impagliazzo, the authors trace what is known, state what is conjectured, and leave the reader at the exact place where the next theorem must be proved.

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P = RP

A Monte Carlo Program for P versus NP

Vertical likelihood, counting N queens, and the complexity of randomness



Nick Polson · Vadim Sokolov

P = RP

Polson and Sokolov